

8 (a) (i) - 9 V

(ii) + 9 V (both (i) and (ii) correct for the mark) B1 [1]

(b) × × B1

✓ × B1

✓ ✓ B1 [3]

(no e.c.f. from (a))

(c) (i) cct: thermistor and resistor in series M1

output connections across thermistor A1 [2]

(ii) as temperature decreases, thermistor resistance increases B1

p.d. across thermistor = $R_T / (R + R_T) \times V$ M1

as R_T increases, output increases A1 [3]

9 (a) product of density (of medium) and speed of sound (in medium) B1 [1]